

CALVIN DSOUZA

PROJECTS & TECHNICAL PORTFOLIO

Cybersecurity · Digital Forensics · Applied AI · Systems Engineering · Research

A curated selection of security systems, AI platforms, software products, low-level engineering projects, and active research investigations.

01 — CYBERSECURITY, FORENSICS & SYSTEMS

AXON

Blockchain Behavioral Forensics & Threat Intelligence Platform

GitHub: <https://github.com/Cal2-0/Axon>

Stack: React · TailwindCSS · D3.js · FastAPI · Python · SQLAlchemy · Etherscan · Alchemy RPC · Forta · Groq

AXON is a blockchain investigation platform built around a different approach to wallet intelligence: **analyse what a wallet does rather than simply what a wallet is**. Instead of relying primarily on static address labels, the platform evaluates behavioural patterns across transactions, wallet relationships, liquidity movement and threat intelligence.

Technical Highlights

- Built a **5-layer behavioural forensics engine** analysing transaction rhythms, fan-out topology and economic pass-through behaviour.
- Developed interactive **D3.js force graphs and Sankey visualisations** for tracing intermediary wallets and capital-flow relationships.
- Implemented a **Prosecution / Defense / Judge multi-agent architecture** for adversarial risk analysis and verifiable threat verdicts.
- Integrated cross-platform OSINT enrichment to correlate on-chain activity with external threat intelligence.
- Analyses a curated corpus of **13,847+ threat entities** across BTC, SOL and EVM ecosystems.

Focus: Blockchain Forensics · Threat Intelligence · Graph Analytics · Behavioural Detection · OSINT

Source details:

SentinelAI

AI-Assisted Linux Incident Response & Security Audit Engine

GitHub: <https://github.com/Cal2-0/Sentinel-IP->

Stack: Python · Streamlit · asyncio · Gemini 2.0 Flash · Isolation Forest · AST Analysis · NVD · EPSS

SentinelAI is a multi-agent Linux security auditing system designed to rapidly determine whether a machine exhibits indicators of compromise. It runs independent security analyses across authentication history, package vulnerabilities and installed-code behaviour before consolidating the findings into a single security context.

Technical Highlights

- **ThreatTrace:** applies Isolation Forest to authentication logs, scoring anomalies using timing, failure patterns and IP reputation.
- **VulnRank:** maps installed pip, apt and npm packages against NVD and EPSS intelligence.
- **PacketSpy:** performs AST-level inspection of installed Python packages to identify suspicious or obfuscated behaviour without dynamically executing the code.
- Uses Gemini as an aggregation layer to convert structured findings into a prioritised security report.
- Supports different audit depths, including rapid dependency checks and deeper system-wide analysis.

Focus: Incident Response · Security Automation · Static Analysis · ML Anomaly Detection · Blue Team Engineering

Source details:

SecureCI

AI-Assisted GitHub Actions Security Scanner & Remediation

GitHub: <https://github.com/Danish4h-135/SecureCI>

Stack: Python · YAML Analysis · Gemini · GitHub Actions · OWASP CI/CD Security

SecureCI is a DevSecOps command-line tool that audits GitHub Actions workflows for common CI/CD security weaknesses and can generate security-focused remediation using Gemini.

Technical Highlights

- Detects **unpinned GitHub Actions**, excessive permissions and exposed secrets.
- Identifies poisoned-pipeline patterns such as unsafe `pull_request_target` usage.
- Detects hardcoded credentials and missing global permission restrictions.
- Supports recursive scanning of `.github/workflows`.
- Produces structured findings and JSON output for downstream CI/CD integration.
- Includes an optional AI-assisted remediation mode that rewrites vulnerable workflow configurations.

Focus: DevSecOps · CI/CD Security · Secure Configuration · Security Automation

The supplied portfolio identifies SecureCI separately from the older SecureCLI entry and documents its GitHub Actions security capabilities.

VisionEX

Enterprise Cybersecurity & Zero-Trust Security Suite

Repository: Not listed in the supplied portfolio

Focus: IAM · OAuth2 · RBAC · Secure Sessions · Network Telemetry · Defense-in-Depth

VisionEX is an enterprise-oriented cybersecurity platform built around zero-trust principles, identity management and layered security controls.

Technical Highlights

- Led a **6-member engineering team** through the development lifecycle.
- Designed authentication architecture around OAuth2, secure session management and strict RBAC.
- Implemented continuous monitoring and security boundary controls.
- Addressed OWASP Top 10 application risks alongside network-level intrusion vectors.
- Portfolio documentation records **three external security audits with zero critical vulnerabilities**.

Focus: Enterprise Security · IAM · Application Security · Zero Trust · Security Architecture

NetScope X v2.0

Real-Time Network Intelligence & Security Telemetry Platform

GitHub: <https://github.com/Cal2-0/Projects/tree/main/netscope>

Stack: Python · Flask · Flask-SocketIO · D3.js · Scapy · Nmap · tshark · Linux / WSL

NetScope X is a browser-based network intelligence platform that continuously discovers devices, captures network traffic, visualises topology and identifies suspicious behaviour.

Technical Highlights

- Combines ARP cache analysis, arp-scan, Nmap, passive packet capture and WSL-aware discovery.
- Builds per-device intelligence including IP, MAC, vendor, hostname, OS indicators, ports and risk level.
- Detects **ARP spoofing, port scans, DNS tunnelling patterns, traffic spikes and newly observed devices**.
- Provides live D3 topology, traffic analysis, event timelines, security alerts and deep device scanning.
- Exposes structured API endpoints for network state, traffic, connections and detected anomalies.

Focus: Network Security · Traffic Analysis · Telemetry · Threat Detection · Visual Analytics

NetRecon

Low-Level Raw-Socket C LAN Scanner

Repository: Not listed in the supplied portfolio

Stack: C · Raw Sockets · ICMP · ARP · TCP · Packet Crafting

A low-level networking project focused on understanding LAN discovery below conventional high-level networking abstractions.

Technical Highlights

- Implemented custom packet-crafting routines for active network mapping.
- Used raw sockets in C to work directly with ICMP, ARP and TCP mechanisms.
- Explored device discovery through low-level network interaction rather than relying solely on standard scanning abstractions.
- Demonstrates practical systems knowledge across **C, packet construction and networking protocols.**

Focus: Systems Programming · Network Internals · Packet Crafting · Reconnaissance

Kalera

Hybrid Post-Quantum Encryption & Steganographic Transport

GitHub: <https://github.com/betrayed1996/Kalera>

Stack: ML-KEM-1024 · Classic McEliece · SPHINCS+ · HKDF-SHA3 · Walsh-Hadamard Diffusion · Steganography · FastAPI · React/Vite

Kalera is a research and demonstration-oriented cryptographic architecture combining **hybrid post-quantum key establishment, layered encryption, steganographic transport and post-quantum signatures.**

Technical Highlights

- Combines **ML-KEM-1024 and Classic McEliece** as a hybrid key-establishment design.
- Uses HKDF-SHA3 for session-key derivation.
- Applies keyed Walsh-Hadamard diffusion and a 4D hyperchaotic keystream layer.
- Embeds encrypted payloads into ordinary images using chaos-positioned LSB steganography.
- Uses **SPHINCS+** for post-quantum message authentication.
- Includes a browser-based demonstration interface and deployment configuration.

Focus: Post-Quantum Cryptography · Secure Communication · Steganography · Applied Cryptography

02 — APPLIED AI, COMPUTER VISION & INTELLIGENT SYSTEMS

MassEd.ex

Real-Time Crowd Intelligence & Event Safety Platform

GitHub: <https://github.com/Cal2-0/Projects/tree/main/Massex>

Stack: Python · OpenCV · YOLOv8 · ByteTrack / DeepSORT · NumPy · Flask

MassEd.ex applies computer vision to real-time crowd safety, transforming live video into actionable information about people, density and potentially dangerous zones.

Technical Highlights

- Developed real-time person detection and multi-object tracking for crowded environments.
- Generated dynamic **crowd-density heatmaps.**
- Implemented region-of-interest monitoring for identifying danger zones.
- Designed the system around continuous video telemetry rather than static image classification.

Focus: Computer Vision · Object Tracking · Crowd Analytics · Real-Time AI · Public Safety

VaidikaAI

Multilingual AI Hospital Triage & Workflow System

GitHub: https://github.com/NITHINKR06/Qwerty201_Protothon

Stack: LLMs · Python · Prompt Engineering

VaidikaAI is a rapid-build hospital triage system designed around multi-model AI-assisted risk assessment and conversational patient-processing workflows.

Technical Highlights

- Designed a multi-model approach for symptom analysis and risk routing.
- Built conversational workflows intended to accelerate patient processing.

- Developed under a **48-hour project constraint**, requiring rapid architecture, integration and delivery.

Focus: Applied AI · Healthcare Workflows · LLM Systems · Rapid Prototyping

OuchMyBrain.io

AI Study Companion — 2nd Place, ACEathon

GitHub: <https://github.com/ACEathon-2025/Team-39>

Stack: Python · Flask · OpenAI / OpenRouter · OCR · TailwindCSS · ElevenLabs

An adaptive learning platform that converts PDFs, scanned notes and images into structured learning material including summaries, flashcards, quizzes and audio lessons.

Technical Highlights

- Built document ingestion and OCR workflows for unstructured study material.
- Developed layered summarisation and key-concept extraction.
- Implemented **Professor Mode** for contextual Q&A and adaptive audio lessons.
- Designed around transforming raw study material into personalised revision workflows.

Recognition: 2nd Place — ACEathon

Focus: Generative AI · OCR · EdTech · LLM Applications · Adaptive Learning

FinTera

Financial Planning & Literacy Platform

Repository: Not listed in the supplied portfolio

Stack: Financial Analytics · Interactive Calculators · Web Application

FinTera is a responsive financial dashboard focused on practical personal-finance planning and decision support.

Technical Highlights

- Built interactive **SIP, EMI and tax calculators**.
- Designed dashboard-based financial planning workflows.
- Focused on making financial calculations and planning more accessible through interactive interfaces.

Focus: FinTech · Financial Analytics · Product Development

03 — ACTIVE RESEARCH & EXPERIMENTAL SYSTEMS

Research projects are intentionally separated from completed software.

These represent active investigations, prototypes and research architectures. Experimental hypotheses are not presented as established results.

Lucent.ai

Interpretable Deepfake Forensics & Synthetic Media Detection

Status: Active Research / In Development

Repository: No public repository

Stack: PyTorch · OpenCV · FFT · Diffusion Reconstruction · FastAPI · Redis · Docker

Lucent.ai investigates whether **mathematically interpretable signal anomalies** can improve the detection of synthetic media across deepfake generation methods that were not seen during development.

Research Question

Which mathematically interpretable signal anomalies are most effective for distinguishing genuine media from deepfakes across unseen generation methods, and does combining complementary anomaly descriptors improve generalization?

Current Research Directions

- Frequency-domain anomaly analysis using FFT.
- Diffusion reconstruction error as a forensic signal.
- Spatial manipulation artefacts.
- Metadata, encoding and camera-signature inconsistencies.
- Fusion of complementary anomaly descriptors.
- Evaluation of cross-generator generalisation.

The emphasis is on understanding **why a media sample appears synthetic**, rather than

treating detection as an opaque binary classifier.

Focus: Deepfake Forensics · Signal Processing · Computer Vision · Anomaly Detection · Generalisation

Kenshō — ケンショウ

Tacit Knowledge Capture & Enterprise Workflow AI

Status: Active Research / System Design

Repository: No public repository

Stack: Python · Django / Flask · PostgreSQL · LLaMA · Elasticsearch

Kenshō explores how organisations can preserve the **practical knowledge that experienced employees carry but rarely document explicitly**.

Rather than functioning as a conventional chatbot, the concept investigates how expert workflows can be captured, structured and transformed into reusable organisational knowledge.

Research Direction

- Capture and structure complex expert workflows.
- Convert tacit procedures into searchable operational knowledge.
- Provide contextual assistance based on captured organisational expertise.
- Explore privacy-first, offline and enterprise deployment models.

Focus: Enterprise AI · Knowledge Systems · Human-AI Interaction · Workflow Intelligence

NeuroMetric

Multimodal Behavioural Analytics Research Platform

Status: Research Prototype / Active Development

Repository: No public repository

Stack: React Native · MediaPipe · Librosa · ClinicalBERT · TensorFlow Lite

An experimental multimodal AI system investigating how **visual, acoustic and language signals** can be combined into structured behavioural representations.

Research Directions

- Gaze stability and visual behavioural signals.
- Facial behavioural features.
- Speech and acoustic characteristics.
- Psychomotor indicators.
- Multimodal signal fusion.
- Privacy-conscious and edge-oriented processing.

Focus: Multimodal AI · Behavioural Analytics · Edge ML · Signal Processing

04 — PRODUCT ENGINEERING & SOFTWARE SYSTEMS

Lyra

Developer Tools & Resources Directory

GitHub: <https://github.com/Cal2-0/guide>

Stack: React 19 · Vite · Tailwind CSS · Radix UI · Framer Motion · Fuse.js

A curated developer-resource platform containing **640+ developer tools, frameworks, databases and APIs**, designed around fast discovery and opinionated recommendations.

Technical Highlights

- Built semantic fuzzy search using Fuse.js.
- Organised resources around developer intent rather than only conventional categories.
- Added persistent **My Stack** functionality through local storage.
- Implemented keyboard-first navigation with ⌘K search.
- Designed a polished application-style interface using React, Tailwind and Framer Motion.

Focus: Product Engineering · Search UX · React · Frontend Architecture

VibeAlchemy

Context-Aware AI Movie Recommendation Extension

GitHub: <https://github.com/Cal2-0/VibeAlchemy>

Stack: Chrome Extension · LLM Semantic Inference · Web Context · Movie APIs

A browser extension that recommends films from natural-language descriptions such as moods, themes and abstract “vibes”, while also using the active webpage as contextual input.

Technical Highlights

- Natural-language semantic recommendation.
- Context-aware matching from pages such as Netflix, Prime Video and IMDb.
- “Surprise Me” and infinite-discovery flows.
- Direct movie discovery from recommendation results.

Focus: Browser Extensions · Recommendation Systems · LLM Applications · Product UX

05 — ENGINEERING LAB

A collection of smaller builds and focused experiments demonstrating breadth across software engineering, AI, security and product development.

Project	What it demonstrates
CalHive	NLP-powered productivity SPA with natural-language task classification using spaCy.
Warehouse Inventory System	Full-stack inventory management covering stock movement, RBAC, capacity tracking and reporting.
The Health Compass	AI health dashboard combining nutrition tracking with TensorFlow-powered food recognition.
Movieszbt	Authenticated movie catalogue and personal watchlist integrating TMDB.
Secure Rooms Chat	Password-protected room-based communication using WebSocket broadcasting.
Universal Studio & Pic Editor	Browser utility combining media metadata extraction with programmatic image manipulation.
Melkit	Digital-forensics utility for extracting and visualising EXIF metadata and GPS coordinates.
ai.co	Modular Flask application acting as a central interface for multiple hosted LLM endpoints.
SnapShop	Image-based product recommendation using CLIP vision-language embeddings and similarity search.
Cipher Lab	Interactive environment for experimenting with classical cryptographic algorithms including Caesar, Vigenère and ROT13.